Fundamentals of Artificial Intelligence and Knowledge Representation
Module-3

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Objective and domain

The domain:

- Professional tennis is a complex, non-linear system.
- A player's **Status** (Ranking, Points) is a strong indicator but doesn't fully capture variables like momentary form or "matchup" compatibility.

The problem:

- Traditional predictive models often focus on *who* will win, failing to explain *why*.
- Ranking alone cannot account for "upsets" (when the underdog wins) or specific in-game dynamics.

Our objective:

To move beyond simple prediction to **diagnostic analysis.**
We use **Bayesian Networks** to model and quantify the causal relationships between pre-match history (*Status*) and actual in-game statistics (*Performance*).

Dataset and Preprocessing

Data Source

- Public **ATP match data** from Jeff Sackmann's GitHub repository.
- Includes match **metadata** (tourney, surface, round) and match **statistics** (aces, service percentage, break points).

Temporal Split

- **Training Set:** 2014 – 2023 (10 years of data to learn dependencies).
- **Test Set:** 2024 (Used to evaluate model performance on unseen data).

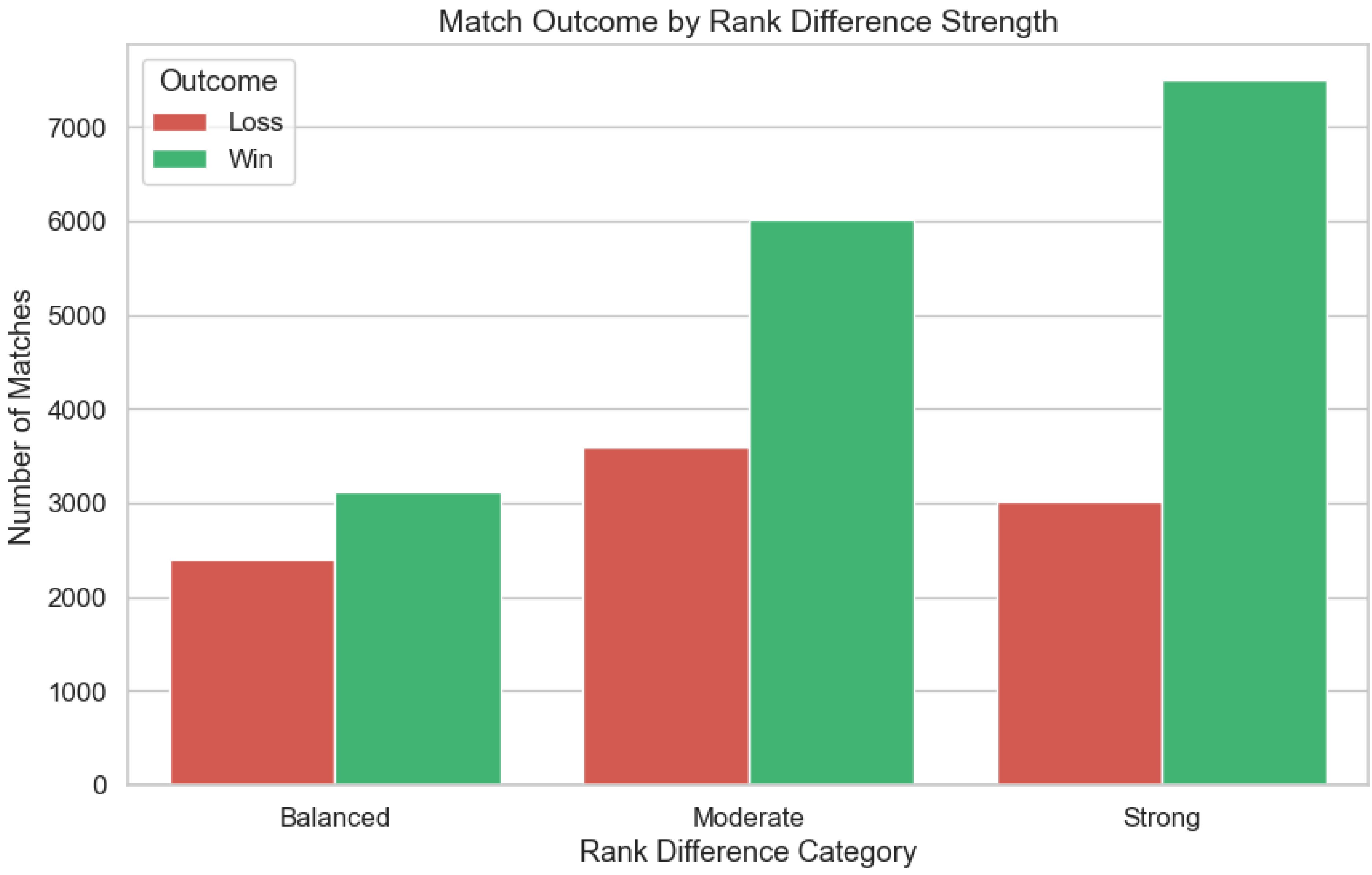
Data Cleaning

- Removed matches with **missing statistics** or incomplete data.
- Excluded Davis Cup matches to focus on standard ATP tournament dynamics.

Discretization

- Bayesian Networks require discrete states.
- Continuous variables (e.g., Ace %, 1st Serve Won %) were converted into 3 bins (**Low, Mid, High**) using **Equal Frequency Discretization**.

Exploratory Data Analysis (EDA)



Main Takeaways

- **Rankings Rule:**
  - Ranking difference is the **strongest** pre-match **predictor** of the outcome.
- **Service is Key:**
  - Aces and 1st Serve Won % are the **clearest** statistics separating winners from losers.
- **Context Matters:**
  - Game dynamics (e.g., Ace frequency) **change** drastically between surfaces (Clay vs. Hard).

Methodology

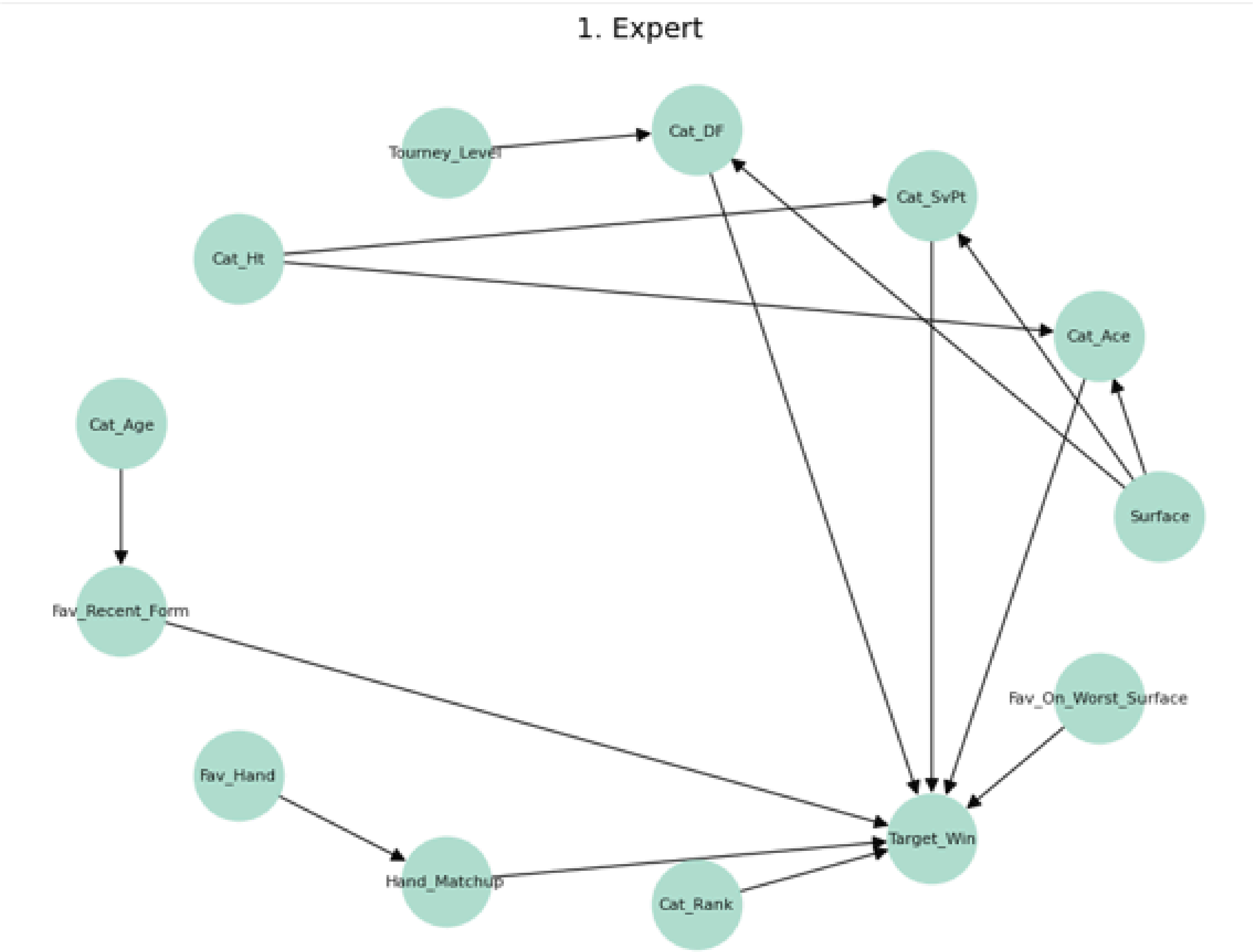
We developed four distinct Bayesian Network architectures using the **pgmpy** library, aiming to shift the focus from mere prediction to diagnostic causal analysis.

The core of our experiment compares two opposing philosophies:

- A **Knowledge-Driven** approach (Expert Model), where the structure is manually designed based on tennis domain logic.
- A **Data-Driven** approach (Learned Model), where the structure is automatically discovered from historical data using the **Hill Climbing** search algorithm.

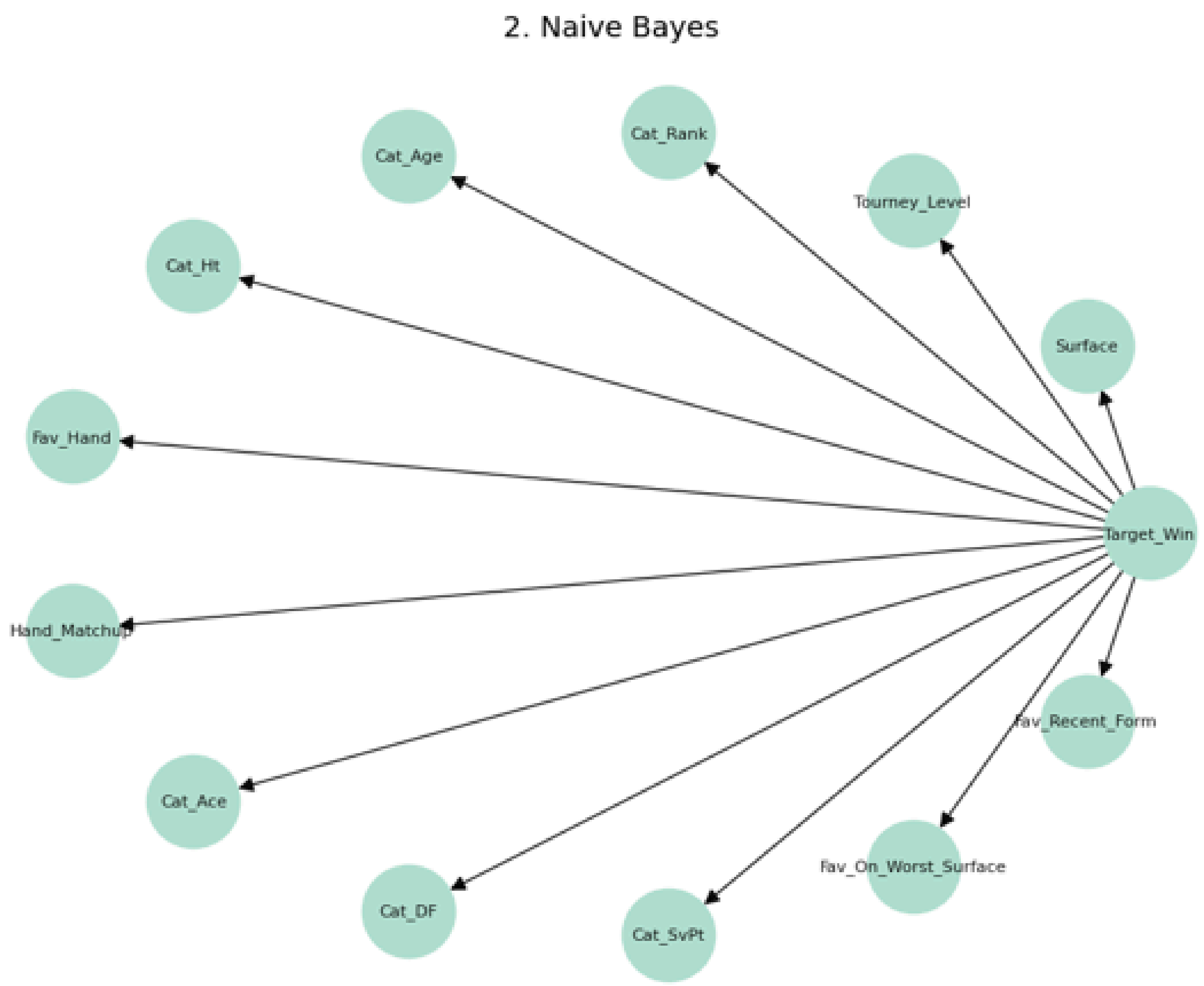
This framework allows us to evaluate whether human domain expertise yields more efficient and interpretable models compared to purely automated algorithms.

1 - Expert Model



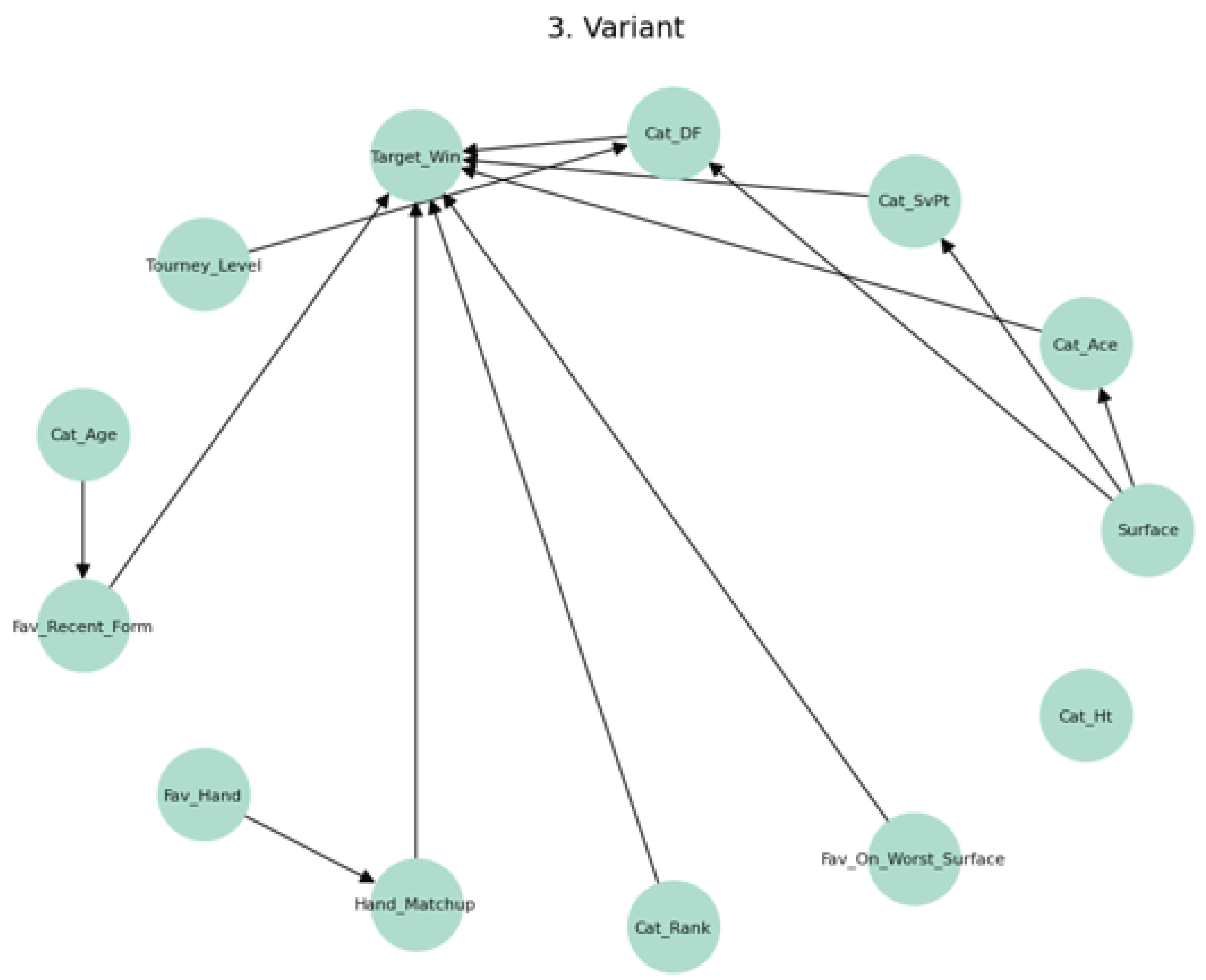
The **key concept** is the use of **human domain knowledge** to define the network structure. Cause-and-effect relationships (such as the impact of the surface on game statistics) are **established manually** based on the physical and tactical laws of tennis, rather than being learned from data.

2 - Naive Bayes Model



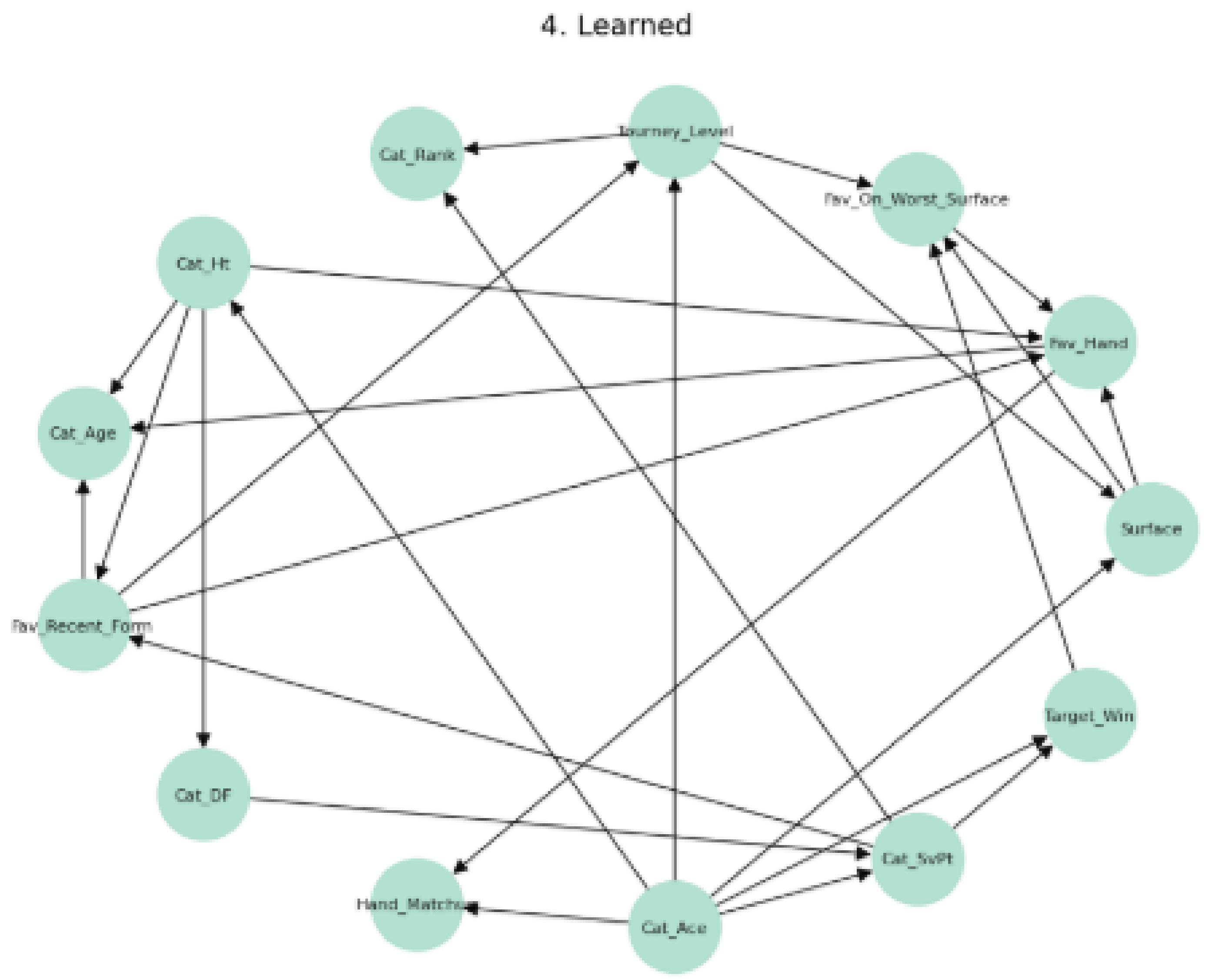
It is based on the statistical assumption of **conditional independence**. The concept is that each match feature (ranking, surface, statistics) influences the result independently of the others. It serves as a **simple baseline** to evaluate whether modeling complex dependencies actually brings an advantage.

3 - Variant Model



It is a **simplified version** of the Expert Model designed to test the hypothesis that certain **physical variables** (like height) are **redundant** when performance data (like Aces) are already present, verifying if a leaner structure maintains the same efficacy.

4 - Learned Model



It is founded on a purely **data-driven approach**. The concept is to ignore any a priori knowledge about tennis and let an algorithm (Hill Climbing) discover the **optimal probabilistic dependencies** based exclusively on the statistical regularities found in the historical data.

Performance evaluation

The inclusion of ex-post performance variables yields **high diagnostic accuracy** on the 2024 test set. The data-driven Learned Model (91.51%) marginally **outperforms** the Expert Model (91.37%), while Naive Bayes lags (90.01%), confirming that tennis variables are **highly dependent** and cannot be treated in isolation.

MODEL	ACCURACY(2024)
1.Expert	91.37%
2.Variant	91.37%
3.Naive Bayes	90.01%
4. Learned (Hill Climbing)	91.51%

Diagnostic Scenarios

1 - The Favourite's Nightmare

We queried the probability of a ”Strong Favorite” winning when forced to play on their worst surface.

- **Result**: The win probability drops to 64.0% (from a baseline > 90%).
- **Diagnosis**: The model correctly identifies that contextual disadvantage (Surface) can neutralize Ranking superiority.

MODEL	Win Probability
1.Expert	64.0%
2.Variant	64.1%
3.Naive Bayes	66.4%
4. Learned	70.5%

Diagnostic Scenarios

2- Service Impact

We measured the sensitivity of the outcome to Service Points Won.

- **Result**: The probability swings from 99.8% (High Service) to 13.2% (Low Service).
- **Diagnosis**: In-match performance acts as a ”switch” that overrides pre-match status, confirming the model’s diagnostic validity

MODEL	High Service	Low Service
1.Expert	99.8%	13.2%
2.Variant	99.8%	13.3%
3.Naive Bayes	99.8%	12.2%
4. Learned	99.8%	9.4%

Diagnostic Scenarios

3 - Pressure Factor

We compared win probabilities for a balanced match in a Grand Slam vs a standard tournament to test stability.

- **Result:** The Expert Model shows high stability (Δ + 0.1%), basing prediction on physical matchups. In contrast, Naive Bayes overestimates the tournament impact (Δ + 10.0%).
- **Diagnosis:** The Expert model is more robust against ”tournament bias,” focusing on the technical reality of the match rather than the event label.

MODEL	Slam Prob	Atp Prob	Delta
1.Expert	64.99%	64.91%	+0.08%
2.Variant	65.09%	65.01%	+0.08%
3.Naive Bayes	64.79%	54.91%	+9.88%
4. Learned	54.10%	57.01%	-2.92%

Conclusion

This study validates Bayesian Networks as a robust diagnostic tool, successfully transforming the problem from simple winner prediction to a causal analysis of match dynamics.

Our key finding is that a **Domain-Driven (Expert)** approach is preferable to a purely **Data-Driven** one. The Expert model matched the accuracy of complex automated algorithms (Learned) while remaining far more interpretable and computationally efficient.

Ultimately, we demonstrated that while *Status* (ranking) sets the expectation, *Performance* (in-match statistics) acts as the causal switch that determines the outcome.

Thank You